

Climate Phenomenon in St. John's [2000-2025]

CMSC6950 Project Proposal

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Data is from ClimateData.ca. ClimateData.ca is a Canadian online portal that provides high-resolution climate data and tools for adaptation planning. The site offers historical and projected data, maps, graphs, and indicators to help users analyze how climate change affects different regions. ClimateData.ca is listed on climate.weather.gc.ca as an option to view station locations on a map and download daily data from multiple stations in bulk in either CSV or GeoJSON format.

The dataset for this project contains daily temporal data for St. John's Area. The data included measurements from four climate observation stations - St. John's West CDA CS, St. John's West Climate, Logy Bay and Signal Hill. The dataset has 36 features with descriptors like "Station_Name", "ID" and "Climate_Identifier" to relevant numeric features like "Mean_Temperature", "Total_Snow" and "Total_Precipitation". The dataset has 10126 examples.

I will analyze Diurnal Temperature Range (DTR), defined as $DTR(t) = T_{\max}(t) - T_{\min}(t)$, which quantifies the daily thermal amplitude. This metric is meaningful because DTR reflects atmospheric stability, cloud cover, and air mass characteristics. For instance, high DTR indicates clear skies and continental influence, while low DTR suggests cloudy conditions. Globally,

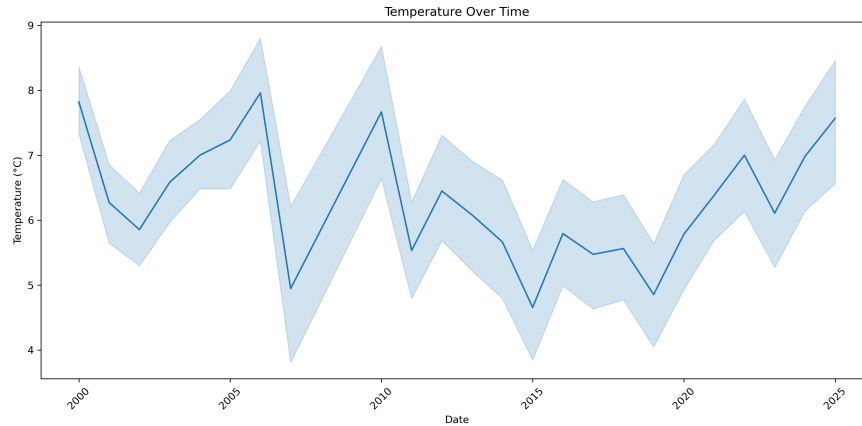


Figure 1: Trendline Mean Temperature in St John's from 2000 to 2025

DTR has been decreasing due to asymmetric warming. In St. John's, I expect winter DTR to be suppressed due to oceanic moderation, while summer DTR should be larger under clearer conditions. If warming is occurring during this period, I anticipate a declining trend in DTR, particularly in winter months when the Arctic effects are strongest.

I will identify extreme DTR days as those exceeding the 90th percentile of the historical distribution and below the 10th percentile, then count the annual frequency of these extremes. To test sensitivity to threshold definitions, I can compare fixed percentiles computed from the entire dataset and seasonal percentiles computed separately for each month to account for baseline differences. This will reveal whether extremes are better defined by climatological context or absolute magnitude.

I will use statistical trend tests appropriate for this time series data to assess whether DTR shows systematic changes over 2000–2025. I test the null hypothesis that no monotonic trend exists, with the alternative being an increasing trend consistent with global warming patterns. Beyond analyzing mean DTR, I will examine whether extreme DTR events (both high and low) show independent trends, addressing whether climate risks are changing in the distribution tails beyond simple mean shifts.